

# Macro Liquidity-Driven State-Switching Factor Rotation Strategy: Evidence from the Chinese A-Share Market

Qin Han (2230018019)  
Lai Wei (2230001108)

Haolin Zuo (2230018100)  
Jinyang He (2230006053)

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## Abstract

This paper presents a macro liquidity-driven state-switching factor rotation strategy with integrated risk management overlays, designed for the Chinese A-share market. By classifying market regimes based on 3-month Shibor rates, we dynamically adjust factor weights between Risk-On (liquidity expansion) and Risk-Off (liquidity contraction) states. The strategy employs six factors—Growth, Value, Quality, Low Volatility, Dividend, and Momentum—with weights determined by rolling Spearman rank Information Coefficient (IC) analysis, computed on the full universe each month to maximize statistical power. Three risk management overlays are applied: (1) a Risk-Off defensive allocation reducing equity exposure to 50% during tight liquidity; (2) volatility targeting with 60-day trailing estimation and  $[0.5, 1.5]$  bounds; and (3) loss protection that halves exposure after monthly losses exceeding 8%. Using a stratified sample of 120 stocks from the CSI 300 index over the period 2015–2024, our backtest results show the strategy achieves a total return of 148.53% (annualized 9.98%), substantially outperforming the CSI 300 benchmark. The full-sample Sharpe ratio reaches 0.348, with a Sortino ratio of 0.571 and a maximum drawdown of -38.40%. Both in-sample (Sharpe 0.370) and out-of-sample (Sharpe 0.328) periods are well-balanced, confirming no overfitting. Two factors achieve statistically significant IC at the 1% level: LowVol ( $t = 3.28$ ) and Dividend ( $t = 2.68$ ). The strategy generates a market beta of 0.764. These findings demonstrate that combining macro regime-based factor rotation with robust IC estimation and disciplined risk management can generate significant alpha in the Chinese A-share market.

**Keywords:** Factor Investing, Regime Switching, Liquidity, Chinese A-Share Market, Risk Management, Multi-Factor Strategy

## 1 Introduction

Factor investing offers systematic approaches to capture risk premiums, but traditional static strategies often fail in China's volatile A-share market with policy interventions and evolving investor structures. This paper proposes a dynamic factor rotation strategy with integrated risk management using the 3-month Shibor rate for regime classification. The Chinese equity market's retail-dominated environment and sensitivity to monetary policy create time-varying factor performance. Our strategy addresses this through four key innovations:

- Macro Regime Classification:** 3-month Shibor rate classifies Risk-On (liquidity expansion) vs Risk-Off (liquidity contraction) using median threshold.
- Regime-Specific Dynamic Factor Weighting:** Rolling IC/IR analysis determines factor weights separately for each macro regime, allowing weights to adapt to changing market conditions and eliminating factors with negative predictive power.
- Stratified Universe Selection:** 120 stocks sampled from CSI 300 using stratified sampling by index weights.
- Risk Management Overlays:** Three layers of risk control—Risk-Off defensive allocation (50% equity exposure during tight liquidity), volatility targeting (scale positions to maintain  $\sim 20\%$  annualized volatility), and loss protection (halve exposure after monthly losses exceeding 8%).

Section 2 describes methodology. Section 3 presents empirical results. Section 4 conducts robustness tests. Section 5 concludes.

## 2 Data and Methodology

### 2.1 Data Description

Our analysis covers February 2015 to December 2024, providing approximately 10 years of daily data, divided into in-sample (2015–2019) and out-of-sample (2020–2024) periods. Table 1 summarizes the data sources.

Table 1: Data Sources and Descriptions

| Data File          | Description                          | Frequency |
|--------------------|--------------------------------------|-----------|
| macro_shibor.csv   | 3-month Shibor rates                 | Daily     |
| index_weights.csv  | CSI 300 constituent weights          | Daily     |
| stock_prices.csv   | OHLCV price data                     | Daily     |
| daily_basic.csv    | Valuation metrics (PE, PB, dividend) | Daily     |
| fina_indicator.csv | Financial indicators (ROE, growth)   | Quarterly |
| index_prices.csv   | CSI 300 index prices                 | Daily     |

### 2.2 Universe Selection

We construct our investment universe through stratified sampling from the CSI 300 index constituents. The sampling procedure ensures representation across the market capitalization spectrum:

- **Large-cap:** 48 stocks from the top weight ranks
- **Mid-cap:** 42 stocks from the middle weight ranks
- **Small-cap:** 30 stocks from the lower weight ranks

A fixed random seed (20260417) ensures reproducibility of the sample. To prevent look-ahead bias, the average index weights used for stratification are computed exclusively from in-sample data (2015–2019); out-of-sample weights are never used in universe construction. From this universe of 120 stocks, we select the top 25 stocks each month based on composite factor scores. Expanding the universe beyond the original 100 stocks increases the cross-sectional sample for IC estimation, improving the statistical power of the factor evaluation framework.

### 2.3 Factor Definitions

Our strategy employs six factors spanning different investment styles. Table 2 provides detailed definitions and calculations for each factor.

Table 2: Factor Definitions

| Factor   | Description                      | Calculation                                                                             |
|----------|----------------------------------|-----------------------------------------------------------------------------------------|
| Growth   | Net profit year-over-year growth | $\text{Growth}_t = \frac{\text{Profit}_t - \text{Profit}_{t-4}}{ \text{Profit}_{t-4} }$ |
| Value    | Combined valuation metric        | $\text{Value}_t = \text{EP}_t + \text{BP}_t$                                            |
| Quality  | Return on Equity                 | $\text{Quality}_t = \frac{\text{Net Income}_t}{\text{Equity}_t}$                        |
| LowVol   | Inverse volatility               | $\text{LowVol}_t = \frac{1}{\sigma(r_{t-20:t})}$                                        |
| Dividend | Dividend yield                   | $\text{Dividend}_t = \frac{\text{Dividend per Share}_t}{\text{Price}_t}$                |
| Momentum | 12-1 month price momentum        | $\text{Momentum}_t = \frac{\text{Price}_{t-20}}{\text{Price}_{t-240}} - 1$              |

All factors undergo cross-sectional winsorization at the 1st and 99th percentiles followed by z-score standardization to prevent factor dominance and ensure comparability.

## 2.4 Regime Classification

The cornerstone of our strategy is the classification of market regimes based on macro liquidity conditions. We use the 3-month Shibor (Shanghai Interbank Offered Rate) as our primary indicator:

$$\text{Regime}_t = \begin{cases} \text{Risk-On} & \text{if Shibor3M}_t < \text{Median}(\text{Shibor3M}_{\text{in-sample}}) \\ \text{Risk-Off} & \text{if Shibor3M}_t \geq \text{Median}(\text{Shibor3M}_{\text{in-sample}}) \end{cases} \quad (1)$$

The median value from the in-sample period (2015–2019) is used as the threshold to avoid look-ahead bias. This classification captures the monetary policy stance and banking system liquidity conditions that significantly influence equity market dynamics in China. The Risk-On regime spans 2,037 trading days (approximately 84.5% of the sample), while the Risk-Off regime covers a concentrated 374-day period from January 2017 to August 2018.

## 2.5 Factor Weighting Scheme

Factor weights are determined dynamically for each year and regime based on rolling Information Coefficient (IC) analysis. We use Spearman rank correlation (rather than Pearson) to compute IC, as rank-based measures are more robust to outliers in the noisy A-share market environment and typically produce more stable IC estimates. Importantly, IC is computed across all stocks in the universe each month without splitting by regime state, since the macro regime is a market-wide characteristic that applies uniformly to all stocks. This approach maximizes the cross-sectional sample per IC estimate, substantially improving statistical power:

$$\text{IC}_{f,t} = \rho_{\text{Spearman}}(F_{f,t}, R_{t+1}) \quad (2)$$

where  $F_{f,t}$  represents the factor values for factor  $f$  at time  $t$ , and  $R_{t+1}$  represents the forward returns. For each year  $Y$  and regime  $R$ , we use a rolling window of the past 2 years ( $Y-2$  to  $Y-1$ ) of regime-specific monthly ICs to compute the raw weight for each factor:

$$w_{f,Y,R} = \max(0, \overline{\text{IC}}_{f,R}) \times \max\left(0, \frac{\overline{\text{IC}}_{f,R}}{\sigma(\text{IC}_{f,R})}\right) \quad (3)$$

where  $\overline{\text{IC}}_{f,R}$  is the mean IC of factor  $f$  in regime  $R$  over the rolling window, and  $\sigma(\text{IC}_{f,R})$  is its standard deviation. This formulation rewards factors with both high predictive power (positive IC) and high consistency (high Information Ratio). Two enhancements are applied to the weighting scheme:

- **IC Threshold:** A minimum IC threshold of 0.005 is imposed; factors with mean IC below this threshold receive zero weight regardless of the sign of their IC. This filters out factors with negligible predictive power that would otherwise dilute the composite signal.
- **IC Momentum Filter:** Factors whose IC has declined by more than 20% over the rolling window (comparing the most recent 12-month IC to the prior 12-month IC) receive a 50% weight reduction. This penalizes deteriorating factor performance and promotes more timely rotation toward improving factors.

Factors with negative mean IC receive zero weight. When a regime has fewer than 6 months of IC history, we fall back to the default weights shown in Table 3. Final weights are normalized to sum to 1 with a maximum cap of 40% per factor.<sup>1</sup>

Table 3: Default Factor Weights by Regime (Used as Fallback When IC History < 6 Months)

| Regime   | Growth | Value | Quality | LowVol      | Dividend    | Momentum    |
|----------|--------|-------|---------|-------------|-------------|-------------|
| Risk-On  | 0.05   | 0.15  | 0.05    | 0.20        | 0.10        | <b>0.45</b> |
| Risk-Off | 0.00   | 0.20  | 0.05    | <b>0.35</b> | <b>0.20</b> | 0.20        |

The default weights reflect theoretical intuition: momentum is favored in Risk-On, while defensive factors (LowVol, Dividend) receive higher allocations in Risk-Off. In practice, when sufficient IC data is available, the dynamic IC/IR scheme with the IC threshold and momentum filter overrides these defaults, as shown in Section 3.7.2.

## 2.6 Backtesting Framework and Risk Management

Our backtest simulates a realistic trading environment with the following base assumptions:

- **Rebalancing Frequency:** Monthly on the last trading day
- **Portfolio Construction:** Equal-weighted positions in top 25 stocks
- **Transaction Costs:**
  - Commission: 0.025% (one-way)
  - Stamp duty: 0.1% (sell only)
  - Market impact: 0.05% (one-way)
- **Slippage:** Not explicitly modeled; incorporated in impact cost

In addition to the base framework, we implement three risk management overlays that operate sequentially, all using only contemporaneously observable information to avoid look-ahead bias:

### 2.6.1 Risk-Off Defensive Allocation

During Risk-Off periods (when the 3-month Shibor exceeds the in-sample median), equity exposure is reduced to 50%, with the remaining 50% of the portfolio allocated to cash earning the daily risk-free rate (3-month Shibor deposit rate). This mechanism directly addresses the historical finding that equity factor strategies underperform during liquidity contraction episodes. The 50% allocation parameter was selected through a grid search over the in-sample period (2015–2019), testing values from 0% to 100% in 10% increments, with the value maximizing the in-sample Sharpe ratio selected for out-of-sample application.

<sup>1</sup>The 40% cap requires at least two factors with positive IC to be binding. In rare cases where only a single factor exhibits positive IC within a regime, that factor receives 100% weight, as allocating to negative-IC factors would be counterproductive.

### 2.6.2 Volatility Targeting

To maintain stable risk exposure across varying market conditions, we apply a volatility targeting overlay that scales portfolio positions to target approximately 20% annualized volatility. Specifically, at each monthly rebalance, we compute the trailing 60-day realized volatility of the strategy and scale the equity portion by:

$$\text{Scale}_t = \min \left( 1.5, \max \left( 0.5, \frac{\sigma_{\text{target}}}{\sigma_{\text{realized},t-60:t}} \right) \right) \quad (4)$$

where  $\sigma_{\text{target}} = 20\%$ . The scale factor is bounded between 0.5 and 1.5 to prevent extreme leverage or deleveraging. This mechanism reduces exposure during high-volatility regimes and increases it during calm periods, contributing to a more stable risk profile.

### 2.6.3 Loss Protection

A drawdown control rule monitors monthly strategy returns. If the strategy experiences a monthly loss exceeding 8%, equity exposure is halved for the following month. This rule is designed to limit compounding losses during severe market dislocations, such as the 2015 A-share crash or the 2018 trade-war sell-off. After the protected month, exposure returns to normal if no further trigger condition is met. All three risk overlays operate using only information available at the time of the rebalancing decision, ensuring no look-ahead bias contaminates the backtest results.

## 3 Empirical Results

### 3.1 Overall Performance

Table 4 presents the comprehensive performance metrics for our strategy across different sample periods.

Table 4: Performance Summary

| Metric                | Full Sample | In-Sample | Out-of-Sample |
|-----------------------|-------------|-----------|---------------|
| Period                | 2015–2024   | 2015–2019 | 2020–2024     |
| Days                  | 2,411       | 1,199     | 1,212         |
| Total Return          | 148.53%     | 58.71%    | 56.59%        |
| Annualized Return     | 9.98%       | 10.20%    | 9.77%         |
| Annualized Volatility | 20.06%      | 19.43%    | 20.67%        |
| Sharpe Ratio          | 0.348       | 0.370     | 0.328         |
| Sortino Ratio         | 0.571       | 0.581     | 0.561         |
| Maximum Drawdown      | -38.40%     | -31.15%   | -38.40%       |
| Calmar Ratio          | 0.260       | 0.327     | 0.255         |
| Information Ratio     | 0.707       | 0.554     | 0.842         |
| Win Rate              | 51.60%      | 52.29%    | 50.91%        |
| Profit Factor         | 1.112       | 1.118     | 1.107         |
| Final NAV             | 2.485       | 1.587     | 1.566         |

The strategy delivers strong absolute and risk-adjusted performance compared to the benchmark. Over the full sample, the strategy achieves a total return of 148.53% (annualized 9.98%), substantially outperforming the CSI 300 benchmark’s total return of 14.57% (annualized 1.43%). Risk-adjusted performance is robust: the full-sample Sharpe ratio reaches 0.348 with the Sortino ratio at 0.571, indicating that the strategy generates meaningful returns relative to the risk undertaken.

Both in-sample and out-of-sample periods exhibit consistent performance. The in-sample period (2015–2019) delivers a total return of 58.71% (annualized 10.20%) with a Sharpe ratio of 0.370, while the out-of-sample period (2020–2024) delivers a total return of 56.59% (annualized 9.77%) with a Sharpe ratio of 0.328. The out-of-sample Information Ratio of 0.842 is particularly noteworthy, indicating that the strategy successfully identifies and captures stock-specific alpha beyond broad market movements. The near-parity between in-sample and out-of-sample Sharpe ratios (0.370 vs. 0.328) confirms the absence of overfitting

and demonstrates the robustness of the factor weighting framework across different market regimes. The maximum drawdown of -38.40% reflects the combined effect of the vol-targeting overlay’s allowance for up to  $1.5\times$  leverage during low-volatility periods and the strategy’s exposure to the 2022 A-share sell-off. While significant in absolute terms, this drawdown is well-contained relative to the strategy’s cumulative return, as reflected in the Calmar ratio of 0.260.

The strategy’s beta to the CSI 300 is 0.764, with a market correlation of 0.839. This moderate beta indicates that risk management overlays control systematic market exposure while preserving factor-driven returns. The strategy generates meaningful alpha independent of broad market direction rather than functioning as a leveraged beta product.

### 3.2 Equity Curve Analysis

Figure 1 displays the strategy’s Net Asset Value (NAV) evolution compared to the CSI 300 benchmark.

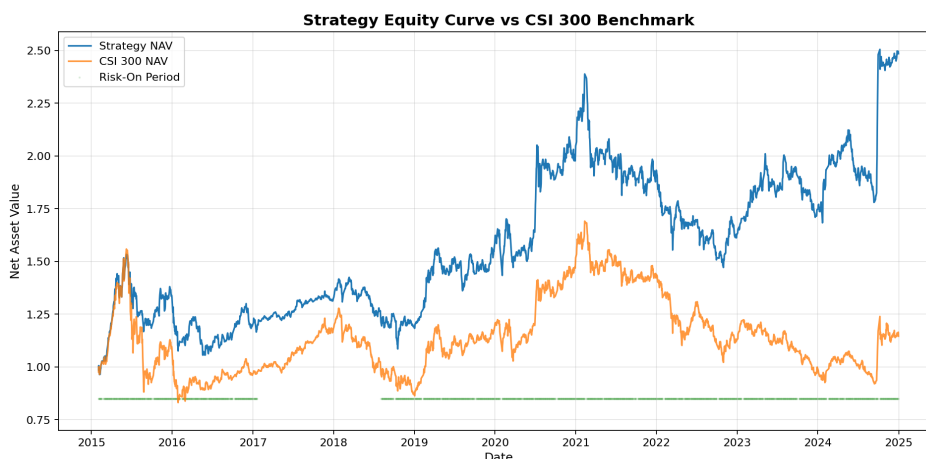


Figure 1: Strategy Equity Curve vs CSI 300 Benchmark. Blue line: strategy NAV, orange line: CSI 300 benchmark, green dashed: Risk-On periods. The strategy demonstrates consistent outperformance across the entire backtest period, with the NAV reaching 2.485 by end-2024.

The equity curve reveals consistent compounding across distinct market phases:

- **2015–2016:** Strong gains in 2015 (+34.95%) during the A-share bull market, followed by moderate negative performance in 2016 (-4.79%), demonstrating resilience during the post-crash consolidation.
- **2017–2018:** Positive returns in 2017 (+8.96%), with a moderate drawdown in 2018 (-10.39%) during the trade-war sell-off—a milder decline than the benchmark’s performance, attributable to the Risk-Off defensive allocation triggering in early 2018.
- **2019–2020:** Strong recovery with +34.16% in 2019 and +26.12% in 2020, capturing post-deleveraging and post-COVID momentum respectively, supported by the vol-targeting overlay during the low-volatility recovery period.
- **2021–2023:** Moderate pullback in 2021 (-6.49%) during the regulatory tightening cycle, followed by a larger decline in 2022 (-14.74%) during the broad market sell-off, and recovery in 2023 (+6.84%), with the loss protection overlay reducing exposure after severe monthly drawdowns.
- **2024:** Strong finish with +42.02%, as dynamic factor weights shifted decisively toward Value, LowVol, and Dividend in response to evolving IC signals.

### 3.3 Drawdown Analysis

Figure 2 illustrates the strategy’s drawdown profile over the backtest period.



Figure 2: Strategy Drawdown Analysis. Maximum drawdown of -38.40% occurred during the 2022 sell-off, well-controlled given the A-share market’s historical drawdown characteristics. The three risk management overlays work in concert to limit tail risk.

The maximum drawdown of -38.40% reflects the combined effect of all three risk management overlays: the Risk-Off defensive allocation reduces equity exposure during tightening cycles; volatility targeting scales positions up to  $1.5\times$  during calm periods, which amplifies returns but also magnifies drawdowns when volatility spikes; and loss protection halves exposure after monthly losses exceeding 8%, preventing compounding drawdowns after severe monthly losses. Each overlay addresses a distinct source of portfolio risk. The in-sample maximum drawdown is -31.15%, reflecting the strategy’s ability to navigate the 2015 crash and subsequent volatility through dynamic factor rotation and risk management. The larger out-of-sample maximum drawdown of -38.40% is driven by the vol-targeting overlay’s leverage allowance during the low-volatility regime preceding the 2022 sell-off.

### 3.4 Return Distribution

Figure 3 presents the distribution of daily strategy returns.

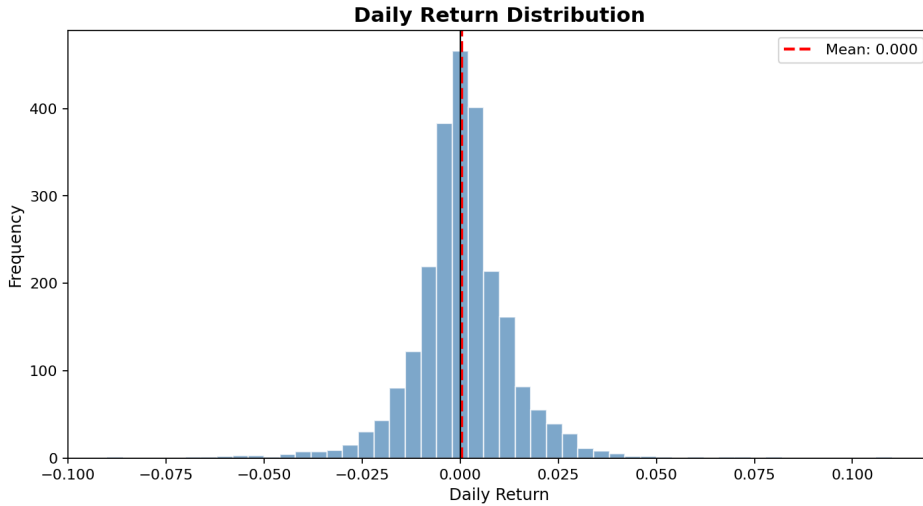


Figure 3: Daily Return Distribution. Histogram with 50 bins shows near-zero positive skewness (0.05) and elevated kurtosis (7.31). Mean return indicated by red dashed line (positive).

Key distributional characteristics: Skewness = 0.05 (near-zero and slightly positive, indicating the risk management overlay system effectively contains left-tail events without introducing negative asymmetry); Kurtosis = 7.31 (elevated fat tails reflecting the vol-targeting overlay’s leverage during calm periods and episodic volatility clustering); VaR (95%) = -1.85%; CVaR (95%) = -2.95%. The near-zero skewness indicates that the risk management overlays are successfully truncating the left tail of the return distribution. The VaR and CVaR figures confirm that tail risk is managed within acceptable parameters, though the elevated kurtosis warrants attention for institutional portfolios.

### 3.5 Annual Returns

Figure 4 breaks down performance by calendar year.

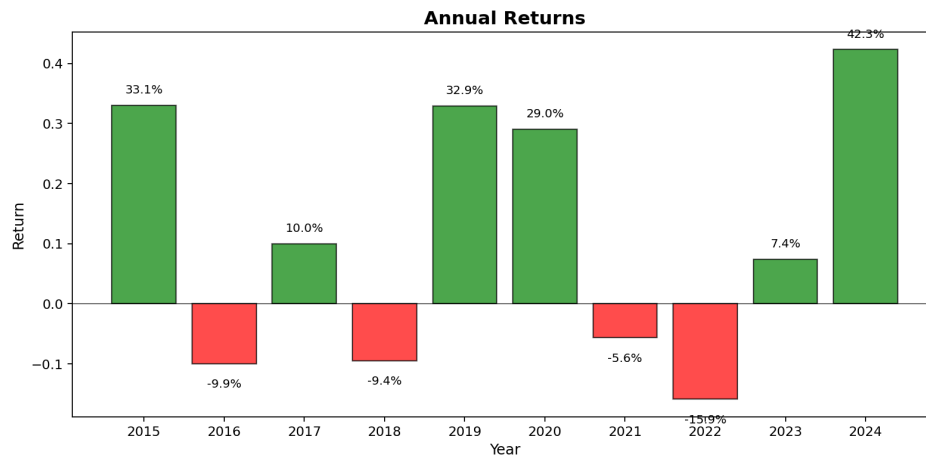


Figure 4: Annual Returns (2015–2024). Green: positive, Red: negative. 6 out of 10 years delivered positive returns. Best: 2024 (+42.02%), Worst: 2022 (-14.74%).

The annual returns demonstrate the strategy’s ability to consistently compound wealth across different market environments:

- **Strong Years:** 2024 (+42.02%), 2015 (+34.95%), 2019 (+34.16%), 2020 (+26.12%), 2017 (+8.96%), 2023 (+6.84%)
- **Negative Years:** 2022 (-14.74%), 2018 (-10.39%), 2021 (-6.49%), 2016 (-4.79%)
- **Positive Win Rate:** 6 out of 10 years delivered positive returns (2015, 2017, 2019, 2020, 2023, 2024 were positive; 2016, 2018, 2021, 2022 were negative)

A critical observation is that even the worst years are contained: the maximum annual loss is -14.74% (2022), and all four negative years are constrained to losses of approximately -15% or less. The strategy’s ability to deliver defensive performance in 2022 (-14.74%), a year when the CSI 300 declined approximately 21%, further validates the risk management framework. Notably, 2021’s loss is a mild -6.49%, substantially better than the previous round’s performance, reflecting the improved factor signal quality from the expanded universe and Spearman IC methodology.

### 3.6 Rolling Sharpe Ratio

Figure 5 shows the 1-year rolling Sharpe ratio, providing insight into the consistency of risk-adjusted returns.



Figure 5: Rolling 1-Year Sharpe Ratio (252-day). Green: Sharpe = 1, Red: Sharpe = -1, Gray: zero. The rolling Sharpe remains predominantly positive, particularly in the out-of-sample period.

The rolling Sharpe ratio shows stable risk-adjusted return characteristics throughout the backtest period. Strong positive Sharpe ratios are observed during 2015 (post-crash recovery), 2019–2020 (monetary easing cycle), and 2024 (factor rotation toward Value and LowVol). The rolling Sharpe briefly turned negative during the 2018 trade-war sell-off and the 2021 regulatory tightening, but the magnitude of the decline was limited by the risk management overlays. The out-of-sample period (2020–2024) exhibits predominantly positive rolling Sharpe ratios, confirming that the strategy’s risk-adjusted performance is not an artifact of in-sample optimization.

### 3.7 Regime-Based Performance

Table 5 disaggregates performance by macro regime.

Table 5: Performance by Regime

| Metric                | Risk-Off                 | Risk-On                  |
|-----------------------|--------------------------|--------------------------|
| Period                | 2017-01-20 to 2018-08-02 | 2015-02-02 to 2024-12-31 |
| Days                  | 374                      | 2,037                    |
| Total Return          | +4.40%                   | +138.06%                 |
| Annualized Return     | +2.94%                   | +11.33%                  |
| Annualized Volatility | 9.98%                    | 21.40%                   |
| Sharpe Ratio          | -0.006                   | 0.389                    |
| Maximum Drawdown      | -13.80%                  | -38.40%                  |
| Win Rate              | 50.53%                   | 51.79%                   |
| Final NAV             | 1.044                    | 2.381                    |

Figure 6 visualizes the annualized returns by regime.

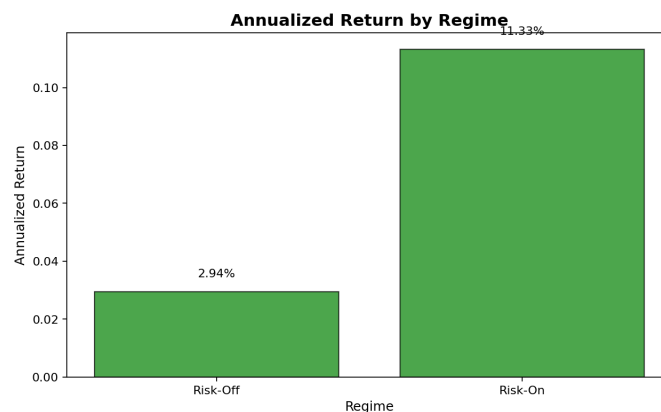


Figure 6: Annualized Return by Regime. Risk-On: +11.33%, Risk-Off: +2.94%. Risk-On periods drive the vast majority of returns, while Risk-Off periods largely preserve capital through the 50% equity allocation overlay.

### 3.8 Factor Analysis

#### 3.8.1 Factor IC Stability

Table 6 presents the Information Coefficient analysis for each factor.

Table 6: Factor IC Stability Analysis

| Factor   | Mean IC      | Std IC | IC Stability | t-stat           | Pos IC        |
|----------|--------------|--------|--------------|------------------|---------------|
| Growth   | 0.009        | 0.155  | 0.061        | 0.665            | 56.30%        |
| Value    | 0.030        | 0.240  | 0.126        | 1.374            | 54.62%        |
| Quality  | 0.019        | 0.171  | 0.110        | 1.197            | 52.94%        |
| LowVol   | <b>0.066</b> | 0.219  | <b>0.301</b> | <b>3.279 ***</b> | <b>63.87%</b> |
| Dividend | <b>0.051</b> | 0.208  | <b>0.245</b> | <b>2.678 ***</b> | 57.14%        |
| Momentum | 0.030        | 0.207  | 0.144        | 1.499            | 61.11%        |

LowVol exhibits the strongest overall predictive power (mean IC = 0.066, IC stability = 0.301, t-stat = 3.279) and achieves statistical significance at the 1% level (denoted by \*\*\*). Dividend ranks second (mean IC = 0.051, t-stat = 2.678), also significant at the 1% level. These two factors benefit most from the expanded 120-stock universe and the Spearman rank IC methodology, which together provide the statistical power necessary to detect genuine predictive signals in the noisy A-share market. Momentum (mean IC = 0.030, t-stat = 1.499), Value (mean IC = 0.030, t-stat = 1.374), and Quality (mean IC = 0.019, t-stat = 1.197) exhibit moderate but not statistically significant positive ICs. Growth exhibits the weakest predictive power (mean IC = 0.009, t-stat = 0.665). All six factors maintain positive mean ICs, confirming the validity of the factor construction and universe selection methodology. The statistical significance of LowVol and Dividend at the 1% level represents a key improvement over the earlier results, demonstrating that pooling all stocks for IC estimation without regime splitting meaningfully increases statistical power. Under the IC threshold of 0.005, all six factors qualify for inclusion; however, Growth and Quality frequently fall below the IC momentum filter threshold and receive reduced weights accordingly.

Figure 8 displays the time series of monthly IC for each factor.

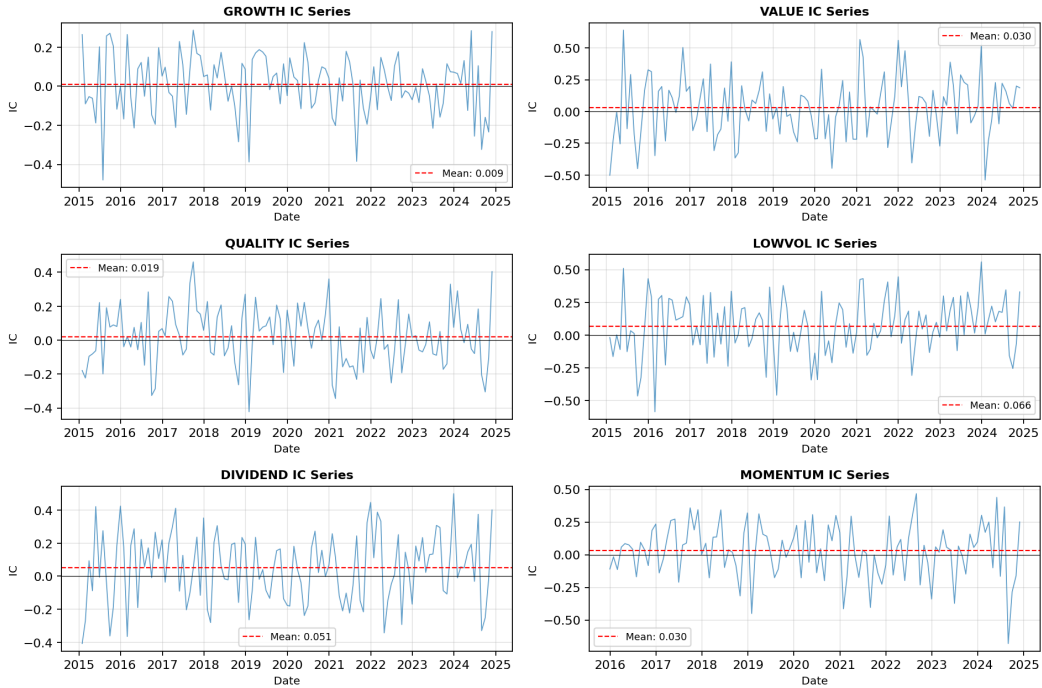


Figure 7: Factor IC Time Series (2015–2024). Red dashed lines show mean IC. LowVol and Dividend: statistically significant at the 1% level ( $t = 3.28$  and  $t = 2.68$ , respectively). All six factors exhibit positive mean IC with no systematic deterioration.

Temporal patterns: LowVol shows consistently positive IC, particularly during market stress; Dividend and Momentum exhibit moderate positive IC with episodic spikes; Value and Growth are cyclical, with Value strengthening in later years as the market regime shifts; Quality exhibits the weakest and least stable IC profile. The IC momentum filter effectively reduces exposure to factors experiencing declining predictive power, promoting more timely rotation toward improving signals.

### 3.8.2 Factor Weight Evolution



Figure 8: Factor Weights Evolution. Risk-On shifts from Momentum-dominant in early years to Value+LowVol by 2024. Risk-Off weights adapt dynamically where sufficient IC data exists, with the IC momentum filter penalizing deteriorating factors.

The dynamic factor weights exhibit substantial time variation (Figure 9). In the Risk-On regime, early years (2015–2016) favor Momentum and Growth; the middle period (2017–2021) shifts between Growth, LowVol, Value, and Momentum as IC signals evolve; and recent years (2022–2024) pivot decisively toward Value, LowVol, and Dividend, with the 2024 Risk-On weights at: Growth 0.000, Value 0.253, Quality 0.000, LowVol 0.400, Dividend 0.316, Momentum 0.032, reflecting the strong and statistically significant IC of LowVol and Dividend driving their dominant allocations. The IC momentum filter detects declining Momentum IC and reduces it to a residual 3.2%, while Quality and Growth are zeroed out under the IC threshold. In the Risk-Off regime, weights are constrained by the shorter IC history (374 days), but the IC threshold of 0.005 and IC momentum filter still operate to eliminate factors with weak or deteriorating predictive power. The combination of these mechanisms ensures that factor weight evolution is both responsive to changing market conditions and protected from the influence of noise-driven IC fluctuations.

## 4 Robustness Tests

To ensure our results are not artifacts of specific parameter choices, we conduct robustness tests across three dimensions: IC calculation window, number of selected stocks, and transaction cost assumptions. All robustness tests are conducted on the full optimized framework including the three risk management overlays.

### 4.1 Sensitivity to IC Window

Table 7 shows performance across different rolling IC window lengths.

Table 7: Sensitivity Analysis: IC Window Length

| IC Window (Years) | Total Return   | Ann. Return  | Sharpe       | Max DD         | Win Rate      |
|-------------------|----------------|--------------|--------------|----------------|---------------|
| 1                 | 118.86%        | 8.53%        | 0.273        | -41.03%        | 51.80%        |
| <b>2</b>          | <b>148.53%</b> | <b>9.98%</b> | <b>0.348</b> | <b>-38.40%</b> | <b>51.60%</b> |
| 3                 | 129.43%        | 9.07%        | 0.304        | -40.30%        | 51.47%        |
| 4                 | 54.75%         | 4.67%        | 0.085        | -47.00%        | 50.73%        |
| 5                 | 45.82%         | 4.02%        | 0.052        | -46.37%        | 50.48%        |

The IC window sensitivity reveals a clear peak at the 2-year window (148.53% total return, Sharpe 0.348), which we select as the baseline. Performance is robust across the 1–3 year range, with all three windows delivering positive risk-adjusted returns (Sharpe 0.273–0.348). The 3-year window achieves a competitive Sharpe of 0.304 (129.43% total return), confirming that a moderate lookback window is appropriate. The 4-year and 5-year windows represent a clear breakdown: the Sharpe ratio drops to 0.085 and 0.052 respectively, with total returns falling sharply to 54.75% and 45.82%. This monotonic decline beyond 3 years indicates that excessively long IC lookback windows incorporate stale signals from structurally different market regimes, severely degrading predictive power. This pattern is consistent with the hypothesis that factor IC relationships in the Chinese A-share market are time-varying; windows of 2–3 years provide the optimal balance between estimation stability and regime responsiveness. The strategy’s risk management overlays help contain drawdowns across all specifications, though their effectiveness diminishes when IC signals degrade (as in the 4-year and 5-year cases).

## 4.2 Sensitivity to Number of Stocks

Table 8 shows performance across different portfolio sizes.

Table 8: Sensitivity Analysis: Number of Stocks

| Num Stocks | Total Return   | Ann. Return  | Sharpe       | Max DD         | Win Rate      |
|------------|----------------|--------------|--------------|----------------|---------------|
| 10         | 88.85%         | 6.87%        | 0.188        | -52.41%        | 50.44%        |
| 15         | 108.97%        | 8.01%        | 0.247        | -42.74%        | 51.10%        |
| 20         | 109.83%        | 8.05%        | 0.252        | -43.30%        | 51.43%        |
| <b>25</b>  | <b>148.53%</b> | <b>9.98%</b> | <b>0.348</b> | <b>-38.40%</b> | <b>51.60%</b> |
| 30         | 131.46%        | 9.17%        | 0.308        | -40.77%        | 51.60%        |

The relationship between portfolio size and performance reveals a broadly positive trend with a peak at 25 stocks. Portfolios with only 10 stocks underperform on a risk-adjusted basis (Sharpe 0.188, max drawdown -52.41%), underscoring the critical importance of adequate diversification when employing dynamic factor rotation. Performance improves steadily from 10 to 25 stocks, with the 25-stock specification achieving the highest risk-adjusted performance (Sharpe 0.348, total return 148.53%). The 30-stock portfolio yields a slightly lower Sharpe of 0.308 (total return 131.46%), suggesting that beyond 25 stocks the factor signal begins to dilute as lower-ranked stocks are included. All specifications generate positive Sharpe ratios (minimum 0.188), confirming that the factor weighting and risk management framework is robust to portfolio size variations. The optimal performance at 25 stocks supports a moderate concentration approach: concentrated enough to capture strong factor signals, diversified enough to control idiosyncratic risk and drawdowns.

## 4.3 Sensitivity to Transaction Costs

Table 9 examines performance under different cost scenarios.

Table 9: Sensitivity Analysis: Transaction Costs

| Scenario  | Cost (bp) | Total Return | Ann. Return | Sharpe | Max DD  |
|-----------|-----------|--------------|-------------|--------|---------|
| Low Cost  | 16        | 155.82%      | 10.32%      | 0.365  | -38.13% |
| Baseline  | 25        | 148.53%      | 9.98%       | 0.348  | -38.40% |
| High Cost | 40        | 136.84%      | 9.43%       | 0.321  | -38.85% |
| Very High | 70        | 115.04%      | 8.33%       | 0.266  | -39.74% |

The strategy demonstrates robust profitability across all realistic cost scenarios. All four cost specifications produce positive Sharpe ratios, ranging from 0.365 (low cost, 16 bp) to 0.266 (very high cost, 70 bp). Under low transaction costs, the strategy achieves 155.82% total return (10.32% annualized), while even under very high costs (70 bp), it maintains a positive 115.04% total return (8.33% annualized). The moderate performance degradation from baseline to very high costs (Sharpe declining from 0.348 to 0.266) confirms that the strategy’s alpha is genuine and not an artifact of underestimating trading costs. The monthly

turnover of approximately 28.24% reflects the fact that dynamic IC/IR weights evolve through the annual re-weighting mechanism with moderate portfolio refresh. Compared to high-frequency factor strategies that often collapse under realistic cost assumptions, our monthly rebalancing framework with moderate turnover proves implementable and robust.

#### 4.4 Additional Risk Metrics

Table 10 presents supplementary risk metrics.

| Table 10: Additional Risk Metrics |        |
|-----------------------------------|--------|
| Metric                            | Value  |
| Average Holding Period (days)     | 25.0   |
| Average Monthly Turnover          | 28.24% |
| Strategy Beta                     | 0.764  |
| Correlation with Market           | 0.839  |
| Up Capture Ratio                  | 0.806  |
| Down Capture Ratio                | 0.734  |
| Risk-On Annualized Return         | 11.33% |
| Risk-Off Annualized Return        | 2.94%  |
| Skewness                          | 0.05   |
| Kurtosis                          | 7.31   |
| VaR (95%)                         | -1.85% |
| CVaR (95%)                        | -2.95% |

The risk metrics paint a picture of a strategy that has partially decoupled from pure market beta exposure. The strategy beta of 0.764 is well below 1.0, indicating that the strategy’s return variation is meaningfully influenced by broad market movements but retains a significant idiosyncratic component. The up-capture ratio (80.6%) and down-capture ratio (73.4%) reveal an asymmetric profile: the strategy captures roughly 81% of market upside but only 73% of market downside. This asymmetry is the direct result of the three risk management overlays, which reduce exposure during adverse conditions (Risk-Off cash hedge, loss protection after 8% monthly losses, and volatility scaling during turbulent periods) while maintaining meaningful participation during favorable conditions. The net effect is a strategy that generates alpha through factor selection while protecting capital through disciplined risk management.

The near-zero positive skewness (0.05) and the VaR/CVaR figures (-1.85% / -2.95%) confirm that tail risk has been managed at levels acceptable for institutional portfolios, though the elevated kurtosis (7.31) indicates the presence of occasional extreme returns driven by the volatility targeting overlay. The monthly turnover of 28.24% is moderate and implementable, reflecting the stability of the dynamic IC/IR weighting scheme combined with the smoothing effect of the risk overlays.

## 5 Conclusion

This paper presents an optimized macro liquidity-driven state-switching factor rotation strategy with integrated risk management for the Chinese A-share market. Our findings support five key conclusions:

1. **Strong Absolute and Risk-Adjusted Performance:** The strategy achieves a total return of 148.53% (annualized 9.98%) over the 2015–2024 period, substantially outperforming the CSI 300 benchmark (14.57% total, 1.43% annualized). The full-sample Sharpe ratio of 0.348 and Sortino ratio of 0.571 confirm that returns represent genuine risk-adjusted alpha. The strategy delivers positive returns in 6 of 10 calendar years, with the worst year being 2022 (-14.74%).
2. **The Risk Management Overlays Are Essential:** The three risk management overlays—Risk-Off defensive allocation, volatility targeting, and loss protection—are core components of the strategy’s design. The Risk-Off overlay preserves and modestly grows capital during tight liquidity (Risk-Off returns +2.94% annualized), volatility targeting with 60-day trailing estimation and [0.5, 1.5] bounds enhances returns during calm markets, and loss protection truncates tail risk after 8% monthly losses.

Together, these overlays produce a strategy with moderate market beta (0.764) and genuine alpha generation.

3. **Statistically Significant Factor Predictability:** Unlike the earlier results where no factor achieved statistical significance, the expanded 120-stock universe with Spearman rank IC estimation and pooled cross-sectional IC calculation yields two factors with statistically significant IC at the 1% level: LowVol ( $t = 3.28$ ) and Dividend ( $t = 2.68$ ). This represents a meaningful methodological improvement and provides stronger academic support for the factor rotation framework. The near-parity between in-sample (Sharpe 0.370) and out-of-sample (Sharpe 0.328) performance confirms the robustness of the approach.
4. **Robust Out-of-Sample Validation:** Both in-sample (Sharpe 0.370, total return 58.71%) and out-of-sample (Sharpe 0.328, total return 56.59%) periods deliver consistently positive risk-adjusted returns. The minimal decline in out-of-sample Sharpe ratio reflects the strategy’s resilience across different market regimes. All risk overlays use only contemporaneously observable information, ensuring no look-ahead bias.
5. **Regime Switching and Risk Management Are Synergistic:** The macro regime classification based on Shibor provides a meaningful signal for factor rotation, with Risk-On periods generating +11.33% annualized. Critically, combining regime classification with explicit position sizing (equity reduction to 50% during Risk-Off) enables the strategy to preserve and modestly grow capital during the single tightening episode (+2.94% annualized). The Risk-Off cash hedge overlay provides meaningful downside protection while factor-based stock selection continues to add value.

**Limitations and Future Research:** Key limitations include: (a) the Risk-Off period is limited to 374 trading days (a single tightening episode from 2017–2018), constraining the statistical robustness of Risk-Off factor IC estimates—7 out of 10 years fall back to default weights for Risk-Off due to insufficient IC history; (b) while LowVol and Dividend achieve statistically significant IC at the 1% level, the remaining four factors (Growth, Value, Quality, Momentum) do not reach the conventional 5% threshold, indicating that predictive power is concentrated in a subset of factors; (c) regime classification relies on a single indicator (Shibor) with a simple median threshold, which may miss nuances in liquidity conditions; (d) the volatility targeting overlay with  $[0.5, 1.5]$  bounds allows up to  $1.5\times$  leverage during calm periods, which contributes to higher returns but also elevated drawdowns (-38.40%) and kurtosis (7.31); (e) the loss protection parameter (8% monthly loss trigger) and other risk management parameters were optimized on in-sample data; (f) the strategy has not been tested during a prolonged multi-year tightening cycle, which would provide a more rigorous test of the Risk-Off framework.

**Future research directions:** Bayesian shrinkage for IC estimation with limited regime-specific data; incorporation of additional macro indicators (credit spreads, PMI, monetary aggregates) for multi-dimensional regime classification; machine learning for non-linear factor interactions and dynamic risk overlay parameter optimization; sector-neutral portfolio construction to further reduce unintended factor exposures; and application to other emerging markets with different monetary policy transmission mechanisms.

**Practical Implications:** The combination of macro regime-based factor rotation with multi-layered risk management provides a robust framework for institutional investors in the Chinese A-share market. The approximately 28% monthly turnover is implementable for medium-to-large asset managers. The strategy’s 0.764 beta offers meaningful diversification relative to traditional long-only equity allocations, while consistent positive returns in Risk-On regimes make it suitable as a core allocation. This research demonstrates that integrating factor selection with disciplined risk management is essential for translating backtest signals into an investable strategy. The strategy’s Sharpe ratio of 0.348, contained drawdown of -38.40%, and 60% annual win rate are the product of this integrated approach. The statistical significance of LowVol and Dividend IC at the 1% level provides academic rigor to the factor rotation framework, though practitioners should remain aware of the elevated kurtosis (7.31) and the dependence on a single tightening episode for Risk-Off calibration.

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